

Ergodic theorems

Time



Space

If $x \in X$, its evolution is given by its orbit

$$\{T^{\circ n}(x)\}_n.$$

If $x \in X$, its evolution can be inferred from sets in X . If $B \subset X$ with $\mu(B) > 0$, then x must hit B with frequency $\mu(B)$. Thus a partition of X tells you the distribution of $\{T^{\circ n}(x)\}_n$.

$\bigcup_{n \geq 0} T^{-n}(B) = X$ almost everywhere: the pre-evolution of an event contains all events. Thus every event evolves into every other event eventually.



Ergodicity

$x \in X$ is an occurrence in the state space, and $x \in B$ is an event.